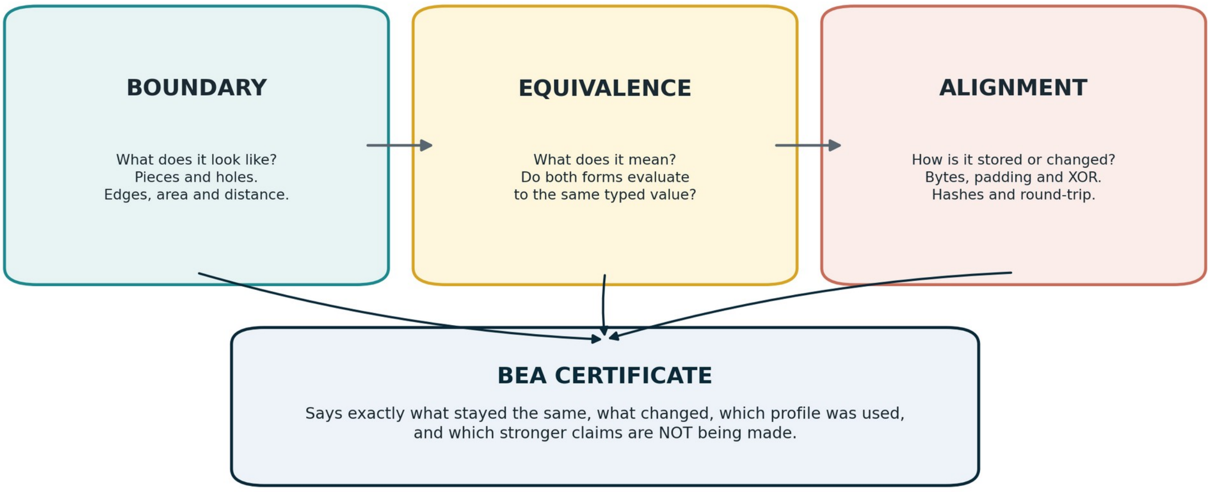
# UGTS-KC 3.6.1 - BEA

Practical Use Cases, Scientific Metrics, and Societal Applications

## Explained simply, without losing the exact engineering meaning

| Field | Value |
|---|---|
| **Prepared for** | Tom Klootwijk |
| **Requester-supplied date of birth** | 10-07-1990 |
| **Requester-supplied identifier** | NL200678942 |
| **Document date** | 17 August 2026 |
| **Evidence rule** | Direct evidence, inherited capability, proposed application and non-claims remain visibly separate. |

### BEA keeps three questions in three different boxes




Simple idea: compare the shape, meaning and code separately - then record the evidence.

**ONE-SENTENCE PRACTICAL MEANING**

Version 3.6.1 is an inspection-and-certificate layer for transformations: it checks shape, meaning and code separately, then records exactly what stayed the same and what changed.

Attribution note: the name, date of birth and identifier above were supplied by the requester and were not independently verified.

# 1. The tiny explanation

Imagine three toy boxes. One box holds the shape of a thing. One box holds what the thing means. One box holds the computer bits used to store it. Two objects can match in one box and differ in the other two. BEA stops us from pretending that one match means everything is identical.

- Boundary asks: what does the representation look like under a declared profile?
- Equivalence asks: do the two representations evaluate to the same typed value?
- Alignment asks: can the exact code change be stated and reversed under a declared byte policy?
- The certificate records preserved features, broken features, evidence level and explicit non-claims.

**`BEA certificate = profiles + boundary signatures + semantic comparison + alignment witness + non-claims`**

The certificate is a structured record. It is not a claim that two shapes are topologically identical.

## The best practical description

BEA is like a careful laboratory clipboard for a transformation. It says: "I changed this representation using these rules; the meaning still matches under this evaluator; the bytes can be checked; these geometric features changed; and I am not claiming anything stronger."

## Where it is most useful

It is most useful when the same real-world item appears in several forms - text, image, sound label, database value, binary message, CAD shape or sensor event - and a silent mix-up between those forms would be expensive, unfair or unsafe.

# 2. What version 3.6.1 has actually established

### What is evidence, and what is only an application idea?

**MEASURED IN VERSION 3.6.1**
Four-font boundary stress data; exact 11-byte XOR round-trip;
bit weights 39/37/38; parity law; 66 passing tests.

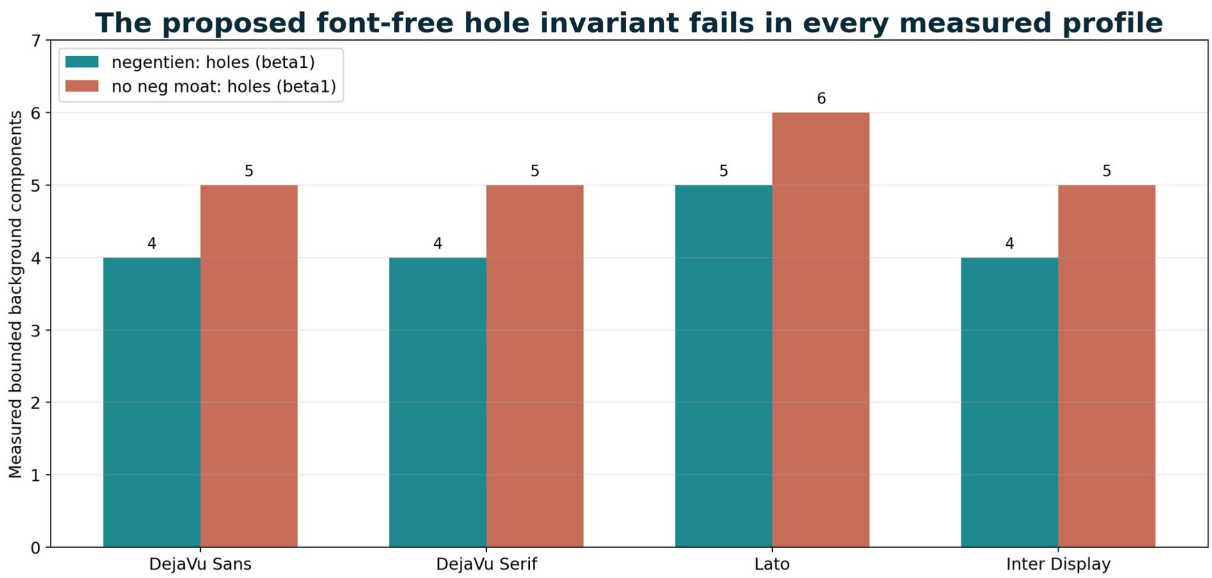
**INHERITED FORMAL UGTS CAPABILITY**
Typed state, support, compatibility, guard, event, transition, lineage,
schemas and bounded hardware metrics.

**PROPOSED ENGINEERING APPLICATIONS**
OCR, data migration, font QA, robotics, CAD, medical-image research,
digital forensics and AI audits.

**NOT ESTABLISHED BY 3.6.1**
Clinical approval, safety certification, linguistic uniqueness, encryption,
universal topology or physical-device advantage.

## Directly measured in the 3.6.1 package

| Measurement | Result | What it means |
|---|---|---|
| **Four-font glyph topology stress test** | The two source phrases do not have the same measured hole count in any of the four profiles. | The claimed font-free topological invariant is not reproduced. |
| **Exact byte alignment witness** | 11-byte left-aligned, NUL-padded ASCII XOR round-trip verifies. | The code delta is exact only under the declared encoding/alignment profile. |
| **Bit weights** | Source 39, target 37, XOR delta 38; parity preserved. | Parity can stay equal while Hamming weight changes. Entropy is not conserved by this fact. |
| **Automated validation** | 66 tests pass. | The bounded implementation is internally consistent across the tested cases. |



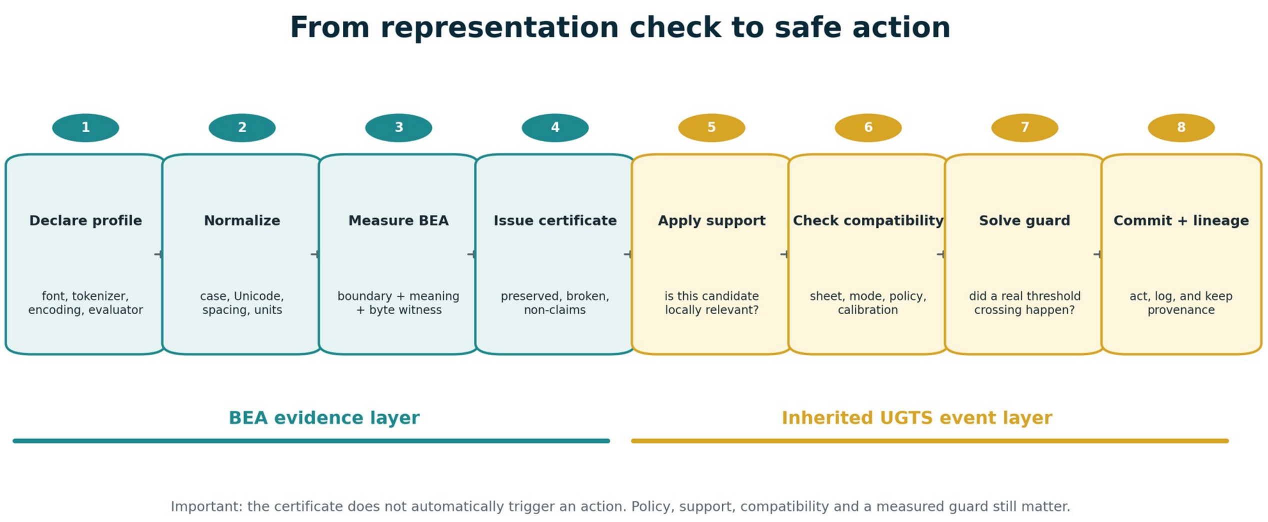

Lesson: geometry and topology must always include the rendering profile.

*Measured beta1 values from the 3.6.1 four-font stress matrix.*

**IMPORTANT BOUNDARY**

The applications later in this guide are engineering uses inferred from the formal machinery. They are not all implemented, benchmarked, clinically validated or safety-certified by version 3.6.1.

# 3. How a real system would use it




## Why the order matters

The operations are not freely interchangeable. Rewriting text before byte encoding is different from editing encoded bytes. Measuring a glyph boundary does not prove semantic equality. A certificate can be valid even when every measured boundary feature changes, provided the declared semantic and alignment checks pass.

**1.** Declare the representation profiles: normalization, tokenizer, renderer, encoder, evaluator and policy.
**2.** Normalize the inputs without silently changing their typed meaning.
**3.** Evaluate the semantic values under declared evaluators.
**4.** Construct the exact code-alignment witness under a declared encoding, padding and alignment rule.
**5.** Measure boundary geometry and topology only when relevant, always with the rendering profile attached.
**6.** Issue a certificate that lists what is preserved, what is broken and what is not claimed.
**7.** For an action system, continue with local support, compatibility, guard crossing, transition and lineage.

**THE PRAGMATIC RULE**

Do not let a representation change trigger a consequential action merely because two values look similar. Require the declared evidence, policy, calibration and event guard.

# 4. Exact metrics: shape, meaning and code

The following measurements turn the child-friendly three-box idea into an engineering test plan. A project should select only the metrics that answer its real failure modes.

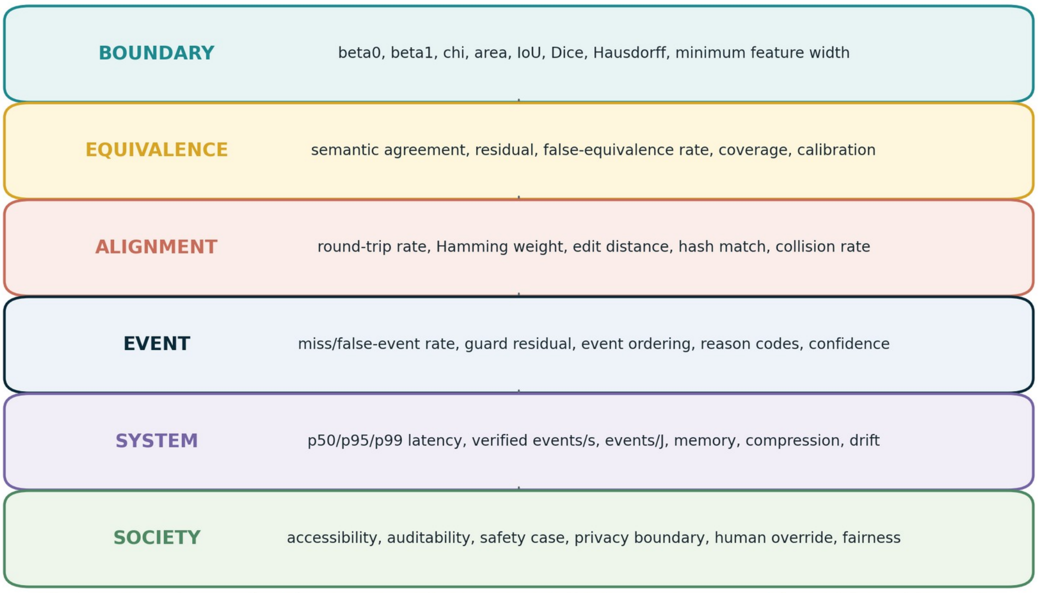
| Metric | Formula or definition | Practical question |
|---|---|---|
| **beta0** | Number of connected foreground components. | Did letters, parts or regions split or merge? |
| **beta1** | Number of bounded background components under declared digital connectivity. | Did a hole or enclosed counter open or close? |
| **chi** | chi = beta0 - beta1. | Did the coarse planar topology signature change? |
| **IoU** | \|A intersect B\| / \|A union B\|. | How much do two masks occupy the same area? |
| **Dice** | 2\|A intersect B\| / (\|A\| + \|B\|). | How similar are two segmentations? |
| **Hausdorff distance** | Worst nearest-boundary distance between two sets. | What is the largest boundary displacement? |
| **Profile robustness** | Passing profiles / profiles tested. | Does the result survive changes of font, encoding or tokenizer? |
| **Semantic agreement rate** | Reference-equal pairs certified equal / reference-equal pairs. | How often does the evaluator preserve intended meaning? |
| **False-equivalence rate** | Reference-different pairs certified equal / reference-different pairs. | How often does the system wrongly say "same"? |
| **Semantic residual** | g_v = \|nu(r) - v\| for a numeric target. | How far is the evaluated value from the target? |
| **Round-trip rate** | decode(encode-transform-inverse) equals original. | Can the transformation be reversed exactly? |
| **Hamming weight/distance** | Number of 1-bits in a vector or XOR delta. | How many bit positions differ? |
| **Hash verification rate** | Matching expected content hashes / checked records. | Did the literal definition or artifact change unexpectedly? |
| **Collision rate** | Distinct inputs mapped to the same output / distinct inputs tested. | Does the transducer erase distinctions? |

**NOT THE SAME METRIC**

A semantic residual of zero is not a glyph signed-distance field of zero. They are zero sets in different domains. Version 3.6.1 was designed specifically to prevent this kind of cross-domain shortcut.

# 5. Exact metrics: events, performance and society

### A practical metric stack




The lower layers do not replace the upper ones. A fast system can still be wrong; a correct model can still be socially unsafe.

| Metric family | Examples | Why it matters |
|---|---|---|
| **Event correctness** | false events, missed events, event ordering, guard residual, reason codes | A correct representation check is useless if the wrong event is committed. |
| **Latency** | p50, p95 and p99 end-to-end latency | The median hides slow tail cases that may dominate safety or user experience. |
| **Throughput** | verified events/s; candidate evaluations/s | Counts useful certified results, not just raw signal propagation. |
| **Efficiency** | verified events/J; full-system energy | Includes source, memory, control, actuation, detection and calibration. |
| **Pruning** | support rejection gain; compatibility rejection gain; event yield | Shows whether cheap gates remove enough irrelevant work. |
| **Memory** | dense bytes, compact event bytes, compression ratio, reconstruction error | Compression is only valid when state remains reconstructible within the error budget. |
| **Stability** | timing jitter, drift, calibration interval, parameter sensitivity | A system that works once but drifts is not deployable. |
| **Social performance** | accessibility pass rate, subgroup error gap, human override rate, audit completeness | Technical correctness alone does not guarantee fair or usable deployment. |

**METRIC DISCIPLINE**

Never report only the easiest number. A fast system can be inaccurate; an accurate system can be unfair; a fair system can still be too slow or too expensive. Report the complete budget relevant to the application.

# 6. OCR, accessible documents and public archives

**LIKE YOU ARE 5**

A camera sees the shapes of letters, a screen reader needs the words, and a computer stores bytes. BEA checks all three so a smudged page does not quietly become the wrong meaning.

## Scientific translation

Use BEA to connect a scanned mask, an OCR token stream and a typed field value. Boundary metrics detect merged or broken glyphs. Equivalence checks the recognized field or numeric value. Alignment records the exact encoding and transformation. A certificate can route uncertain cases to a human instead of silently accepting them.

| Part | What it does | Metrics to report |
|---|---|---|
| **Boundary** | Compare the scan mask and rendered reference; detect closed/open counters and merged characters. | IoU, Dice, Hausdorff, beta0/beta1/chi, minimum stroke and counter width |
| **Equivalence** | Evaluate critical fields such as dates, amounts, identifiers or addresses. | character error rate, word error rate, exact critical-field accuracy, false-equivalence rate |
| **Alignment** | Preserve Unicode normalization, encoding, original bytes and conversion history. | round-trip rate, hash match, collision rate |
| **Operation** | Send low-confidence or profile-breaking cases to review. | coverage, calibration error, p95/p99 latency, human-review rate |

**EXAMPLE ACCEPTANCE RULE**

For a critical field, auto-commit only when the typed value matches exactly, the profile is declared, the confidence is calibrated and no forbidden boundary warning is present. Otherwise hold for review. Numeric thresholds for non-critical text must be set by the document owner.

**SOCIETAL VALUE**

Improves access to archives and forms for screen-reader users, reduces silent digit errors, and creates an auditable reason for human review.

**CAUTION**

Do not treat matching glyph holes as proof that OCR is correct. Text semantics and document context still require a validated evaluator.

| EVIDENCE STATUS | Proposed application built directly from the 3.6.1 boundary/equivalence/alignment machinery; not a shipped OCR engine. |
|---|---|

# 7. Data migration, public records and scientific databases

### LIKE YOU ARE 5

It is like moving the same toy from one box to another. The box can change, but the toy must still be the same toy when you open it.

## Scientific translation

Represent a value in several schemas - for example integer 19, text "19", binary 10011 or a versioned database field. BEA verifies typed semantic equality while recording the exact code conversion and provenance. Boundary checks are optional for visual documents but not needed for pure structured data.

| Part | What it does | Metrics to report |
|---|---|---|
| **Equivalence** | Map source and target records into the same typed value domain. | semantic agreement 100% for required fields; false-equivalence rate 0 in the validated corpus |
| **Alignment** | Verify exact source/target byte policy and reversible transformations. | round-trip 100%; hash verification 100%; untracked transformations 0 |
| **Schema** | Keep units, widths, date conventions, null rules and version identifiers explicit. | schema validation rate, missing-field rate, unit-conversion residual |
| **Lineage** | Record every transformation and external correction. | lineage completeness, replay success, unresolved provenance count |

### EXAMPLE ACCEPTANCE RULE

For a lossless migration, require exact typed equality on every mandatory field, 100% round-trip success where reversibility is promised, zero unknown schema versions and zero unlogged corrections. Any exception must produce a named reason code.

### SOCIETAL VALUE

Supports reliable government records, benefits systems, historical archives and reproducible science by making format changes inspectable rather than magical.

### CAUTION

A content hash proves that bytes match an expected artifact; it does not prove the data are true, private or ethically collected.

| EVIDENCE STATUS | Strong near-term use case. The 3.6.1 package already provides typed evaluators, hashes, schemas, alignment witnesses and lineage concepts. |
|---|---|

# 8. Font, signage and user-interface quality assurance

**LIKE YOU ARE 5**

Changing a font is like drawing the same number with a different crayon. Sometimes a hole closes or two strokes touch, and the symbol becomes harder to read.

## Scientific translation

Render critical glyphs and labels across fonts, sizes, display densities, antialiasing modes and thresholds. Use profile-bound topology and geometry to catch closed counters, disconnected dots, merged digits and unstable icons. Pair these measurements with human readability tests rather than treating topology alone as readability.

| Part | What it does | Metrics to report |
|---|---|---|
| **Topology** | Measure connected pieces and enclosed counters for every declared render profile. | beta0, beta1, chi, topology-change count |
| **Geometry** | Compare masks and boundaries to the approved reference. | IoU, Dice, Hausdorff, area ratio, minimum feature width |
| **Robustness** | Repeat across devices, sizes, weights and raster thresholds. | profile robustness, worst-profile score, failure count |
| **Human use** | Confirm that people can distinguish critical symbols. | confusion matrix, reading time, error rate, accessibility pass rate |

**EXAMPLE ACCEPTANCE RULE**

For safety-critical labels, reject any profile that closes a required counter, merges a critical character pair, falls below the minimum feature width or exceeds the approved human confusion rate. The exact limits must be set by the display and accessibility standard used by the project.

**SOCIETAL VALUE**

Useful for road signs, medical displays, public kiosks, voting interfaces, dashboards and low-vision accessibility testing.

**CAUTION**

beta0, beta1 and chi are diagnostic features, not a complete model of human perception.

| | |
|---|---|
| **EVIDENCE STATUS** | Directly motivated by the four-font 3.6.1 stress test. The shipped test proves profile dependence, not universal readability. |

# 9. Phonetic and multilingual research

**LIKE YOU ARE 5**

Two words can sound a little alike without being the same word. BEA gives the scientist a checklist instead of letting a fun sound-alike turn into a universal law.

## Scientific translation

Treat every pronunciation, spelling, dialect and tokenizer as a versioned profile. Compare phoneme sequences and syllable structures separately from meaning and rendered glyph topology. A claimed alias becomes a bounded hypothesis that can be tested across speakers and dialects.

| Part | What it does | Metrics to report |
| --- | --- | --- |
| **Acoustics** | Compare annotated phoneme or feature sequences. | phoneme edit distance, duration error, acoustic distance |
| **Language** | Check whether independent speakers assign the intended meaning. | human agreement, Cohen-like agreement statistic, false-match rate |
| **Profiles** | Stress dialect, language, speaking rate and recording condition. | cross-profile pass rate, worst-group error, coverage |
| **Representation** | Keep sound, spelling, glyph geometry and numeric meaning separate. | certificate preserved/broken feature counts |

**EXAMPLE ACCEPTANCE RULE**

Promote a phonetic relation only after a preregistered speaker sample, a declared transcription scheme and a false-match analysis. A syllable-count coincidence or matching glyph feature is never enough on its own.

**SOCIETAL VALUE**

Can improve language-learning tools, pronunciation dictionaries and dialect-aware speech interfaces when paired with real linguistic data.

**CAUTION**

Do not infer linguistic uniqueness, cultural truth or semantic equivalence from a constructed sound resemblance.

| | |
| --- | --- |
| **EVIDENCE STATUS** | Version 3.6.1 explicitly leaves the source-proposed Dutch alias unvalidated; the practical contribution is the test framework, not the alias claim. |

# 10. Robotics, sensor fusion and autonomous event decisions

**LIKE YOU ARE 5**

A robot has several teachers: a camera, a distance sensor and a map. It should move only when the teachers are talking about the same nearby thing and the safety line is really crossed.

## Scientific translation

Use BEA to certify that heterogeneous representations refer to the same typed object or hazard. Then use the inherited UGTS sequence: local support removes irrelevant candidates, compatibility checks mode and policy, a guard detects the threshold crossing, and lineage records the decision.

| Part | What it does | Metrics to report |
|---|---|---|
| **Perception** | Connect image regions, point clouds, labels and tracked object IDs. | precision, recall, IoU/Dice, localization error, false-equivalence rate |
| **Support** | Limit work to the local field of view, range and time window. | support rejection gain, candidate count, missed-support rate |
| **Compatibility** | Require sensor mode, calibration, timestamp and policy compatibility. | compatibility rejection gain, reason-code distribution |
| **Event** | Commit only on a verified crossing. | false/missed events, guard residual, event-order error, p99 latency, jitter |
| **Efficiency** | Measure the complete sensing and decision path. | verified events/s, events/J, memory, calibration drift |

**EXAMPLE ACCEPTANCE RULE**

A safety action must have an application-specific hazard budget. Require bounded false-negative and false-positive rates, a maximum p99 latency, calibrated uncertainty and a fallback state. Version 3.6.1 alone is not a safety case.

**SOCIETAL VALUE**

Applicable to warehouse robots, mobility aids, industrial inspection, traffic sensing and assistive navigation.

**CAUTION**

Independent safety engineering, redundancy, fail-safe behavior and domain regulation remain mandatory.

| EVIDENCE STATUS | Proposed application using 3.6.1 certificates plus the inherited UGTS event calculus. |
|---|---|

# 11. CAD, digital twins and manufacturing quality control

**LIKE YOU ARE 5**

A factory can draw the same part as a blueprint, a 3D mesh and a scan of the real object. BEA checks whether they still describe the same part and where the shape changed.

## Scientific translation

Use boundary and topology signatures to compare CAD surfaces, signed-distance fields, meshes and scanned parts. Use semantic equivalence for part identity, units and tolerances. Use lineage to record every design revision, manufacturing step and accepted deviation.

| Part | What it does | Metrics to report |
|---|---|---|
| **Shape** | Compare nominal and measured geometry. | Hausdorff distance, surface residual, volume/area error, tolerance exceedance |
| **Topology** | Detect unintended holes, disconnected regions or merges. | beta0, beta1, chi, component count |
| **Identity** | Keep part, revision and transformation history separate from coordinates. | lineage completeness, revision/hash match |
| **Events** | Detect when deformation or wear crosses a limit. | crossing time, guard residual, false/missed threshold events |
| **System** | Benchmark inspection throughput and uncertainty. | parts/hour, p95 latency, measurement repeatability, calibration interval |

**EXAMPLE ACCEPTANCE RULE**

Accept a part only when every critical tolerance is within the declared uncertainty, no forbidden topology change is detected, the revision hash matches and the inspection lineage is complete.

**SOCIETAL VALUE**

Can reduce scrap, improve maintenance traceability and make digital-twin updates auditable.

**CAUTION**

A coarse topology signature cannot replace full metrology or material testing.

| EVIDENCE STATUS | Proposed application strongly aligned with the boundary, SDF, event and lineage formalism inherited by 3.6.1. |
|---|---|

# 12. Scientific and medical image segmentation research

**LIKE YOU ARE 5**

Two coloring books can shade almost the same organ, but one may accidentally add a hole or miss a tiny island. BEA checks overlap and topology separately.

## Scientific translation

Compare segmentation masks from different algorithms, resolutions or annotation teams. Boundary metrics measure geometric overlap and worst-case displacement. Topology metrics detect component and hole changes. Equivalence records the intended anatomical or scientific label. A certificate exposes disagreements rather than hiding them in one similarity score.

| Part | What it does | Metrics to report |
|---|---|---|
| **Overlap** | Compare predicted and reference masks. | Dice, IoU, precision, recall, sensitivity, specificity |
| **Boundary** | Measure worst and average contour error. | Hausdorff distance, mean surface distance, boundary F-score |
| **Topology** | Detect unwanted islands, tunnels or holes. | beta0, beta1, chi, topology error count |
| **Uncertainty** | Check confidence against observed correctness. | calibration error, coverage, abstention rate |
| **Reproducibility** | Track model, dataset, preprocessing and revision. | hash verification, lineage completeness, rerun agreement |

**EXAMPLE ACCEPTANCE RULE**

No universal Dice threshold is scientifically valid across all tasks. Predefine organ/task-specific geometry, topology and sensitivity limits, validate on an independent cohort, and send uncertain cases to expert review.

**SOCIETAL VALUE**

Encourages transparent medical-imaging research by showing when two masks have similar overlap but different topology.

**CAUTION**

Clinical use requires independent studies, regulatory compliance, cybersecurity, human factors and medical responsibility.

| | |
|---|---|
| **EVIDENCE STATUS** | Proposed research application. Version 3.6.1 is not clinically validated and must not be presented as a diagnostic device. |

# 13. Digital forensics, software patches and archival integrity

### LIKE YOU ARE 5

A detective wants to know exactly which puzzle pieces changed and whether the old puzzle can be rebuilt. XOR can show a change, but it is not a secret lock.

## Scientific translation

Use alignment witnesses, hashes and lineage to record controlled transformations between file versions, protocol messages or archived formats. The 3.6.1 XOR witness is an exact reversible delta under its declared policy. Similar records can support patch verification and chain-of-custody, but cryptographic authenticity requires real cryptographic signatures and key management.

| Part | What it does | Metrics to report |
|---|---|---|
| **Integrity** | Verify expected artifacts and literal definitions. | cryptographic hash match, signature verification, unexpected-change count |
| **Alignment** | Describe exact byte changes and original lengths. | round-trip rate, Hamming distance, delta size |
| **Provenance** | Record who/what transformed the artifact and under which profile. | chain-of-custody completeness, untracked step count |
| **Replay** | Reconstruct prior states from versioned inputs and logs. | replay success, deterministic mismatch count |
| **Performance** | Measure patch and verification cost. | throughput, p95 latency, memory overhead |

### EXAMPLE ACCEPTANCE RULE

For evidentiary or release use, require cryptographic hash/signature verification, 100% reproducible transformation, complete provenance and zero unexplained bytes. Treat plain XOR only as a delta operation, never as encryption.

### SOCIETAL VALUE

Useful for preserving digital evidence, maintaining public archives and making software-update transformations auditable.

### CAUTION

A reversible XOR mask offers no confidentiality and may be trivial to reverse.

| | |
|---|---|
| **EVIDENCE STATUS** | Directly uses the alignment and content-addressing ideas. Cryptographic security is outside the 3.6.1 XOR example. |

# 14. AI robustness, invariance testing and accountable interfaces

### LIKE YOU ARE 5

A smart system should not change its answer just because the same word uses a different font. But it also must not call two different meanings the same. BEA tests both directions.

## Scientific translation

Build profile suites that vary font, encoding, spacing, language, sensor rendering or other representation details while keeping a declared semantic target. Measure whether the model is invariant when it should be and sensitive when it should not collapse distinctions. Store the profile, evaluator, result and non-claims in an audit certificate.

| Part | What it does | Metrics to report |
|---|---|---|
| **Desired invariance** | Same reference meaning across allowed representation changes. | invariance consistency rate, semantic agreement, profile robustness |
| **Forbidden invariance** | Different meanings that the model must distinguish. | false-equivalence rate, collision rate, confusion matrix |
| **Confidence** | Compare confidence to actual correctness and abstain when needed. | calibration error, coverage, selective risk |
| **Fairness** | Repeat across languages, devices and user groups. | subgroup error gap, worst-group performance, profile coverage |
| **Audit** | Preserve prompt/input profile, model version, decision and lineage. | audit completeness, replay agreement, unlogged change count |

### EXAMPLE ACCEPTANCE RULE

Do not accept an invariance claim from one example. Require a declared profile matrix, positive and negative pairs, calibrated confidence, subgroup reporting and reproducible model/version hashes.

### SOCIETAL VALUE

Can make AI testing more transparent by separating representation robustness from semantic correctness and by exposing subgroup failures.

### CAUTION

A BEA certificate does not make a model fair, truthful or safe; it makes a specific test and its limits inspectable.

| | |
|---|---|
| **EVIDENCE STATUS** | Proposed audit harness. The UGTS source explicitly rejects the claim that geometry replaces general AI or model semantics. |

# 15. Smart sensors, edge computing and compact event logs

**LIKE YOU ARE 5**

Instead of drawing every frame forever, a sensor can remember the important moments: "something relevant crossed the line." BEA can certify the representation change before the event system records it.

## Scientific translation

Use profile-bound certificates as preconditions for the inherited UGTS event pipeline. Local support and compatibility prune candidates, guards verify crossings and the system logs only verified novelty where reconstruction is valid. This can reduce output traffic, but only when event density, numerical error and calibration remain bounded.

| Part | What it does | Metrics to report |
|---|---|---|
| **Throughput** | Count candidates and verified events under a fixed profile and error budget. | candidate evaluations/s, verified events/s, event yield |
| **Latency** | Measure end-to-end device and control timing. | p50/p95/p99 latency, timing jitter |
| **Pruning** | Quantify cheap rejection before expensive work. | support rejection gain, compatibility rejection gain |
| **Memory** | Compare dense records with verified-only logs. | state compression ratio, event compaction ratio, reconstruction error |
| **Energy/stability** | Include full system and calibration. | events/J, drift, calibration interval, buffer overflow/missed events |

**EXAMPLE ACCEPTANCE RULE**

Promote an event-only design only when reconstruction is verified, quantization error stays below the guard margin, p99 latency meets the application budget, false/missed events are within limits and the conventional baseline is not better at equal error.

**SOCIETAL VALUE**

Potential uses include environmental monitors, industrial alarms, infrastructure telemetry and low-power edge analytics.

**CAUTION**

Compact event logs lose information unless the unlogged state is truly reconstructible or stored elsewhere.

| | |
|---|---|
| **EVIDENCE STATUS** | Proposed 3.6.1 application with inherited UGTS-GN metrics. The available benchmark is a software Vulkan baseline, not a BEA-specific or physical-GPU result. |

# 16. The inherited performance baseline - and its strict limits

UGTS-GN 1.1 reports a direct Vulkan benchmark for a spherical support/compatibility/guard kernel. It is useful as an example of honest metric naming, but it is not a performance measurement of the new BEA certificate layer.

| Profile/mode | p50 ms | p95 ms | p99 ms | CER M/s | SET M/s | ESB GB/s |
|---|---|---|---|---|---|---|
| **G64_E32 evaluate** | 12.528 | 21.192 | 22.743 | 83.700 | 4.113 | 8.035 |
| **G64_E32 evaluate+commit** | 21.963 | 36.570 | 38.153 | 47.743 | 2.346 | 4.583 |
| **G32_E16 evaluate** | 13.307 | 18.860 | 21.321 | 78.798 | 3.748 | 3.782 |
| **G32_E16 evaluate+commit** | 19.172 | 32.585 | 33.743 | 54.693 | 2.602 | 2.625 |

- Measurement context: N = 1,048,576 candidates; direct Vulkan compute; SwiftShader Device (Subzero), a CPU/software Vulkan device.
- CER means Candidate Evaluation Rate; SET means Spherical Event Throughput; ESB means Effective Substrate Bandwidth.
- The run demonstrates API-native software execution and internal validation on the named device.
- It does not establish physical GPU, FPGA, ASIC, photonic or 3.6.1 BEA throughput.
- A real BEA benchmark must additionally measure profile normalization, semantic evaluation, boundary extraction, certificate construction and storage.

**WHY THIS MATTERS**

Good engineering reports the device, data size, precision, mode, error budget and tail latency. A big throughput number without those conditions is not portable evidence.

# 17. Societal deployment: what responsible use looks like

The strongest societal value of 3.6.1 is not a new gadget. It is disciplined transparency: every transformation states its profile, evaluator, evidence, errors, provenance and non-claims.

| Societal goal | Practical BEA rule | Metric |
|---|---|---|
| **Accessibility** | Stress fonts, devices, languages and assistive outputs; route failures to review. | accessibility pass rate, user error rate, worst-profile score |
| **Fairness** | Do not validate only the easiest profile or dominant user group. | subgroup error gap, worst-group performance, profile coverage |
| **Human control** | Consequential low-confidence cases must allow review or safe fallback. | override availability, review rate, unsafe auto-commit count |
| **Auditability** | Keep profiles, schemas, hashes, reason codes and lineage. | audit completeness, replay success, unlogged change count |
| **Privacy** | Store only the evidence needed; separate identity from geometry; protect raw data. | data retained, re-identification risk assessment, access-log completeness |
| **Accountability** | Name the owner of the evaluator, thresholds and update policy. | version ownership, unresolved exception count, correction time |
| **Scientific honesty** | Publish negative results and counterexamples; keep non-claims in the certificate. | profile failures disclosed, preregistered metrics completed |

**HIGH-STAKES RULE**

In medicine, public safety, benefits, justice, employment or critical infrastructure, a BEA certificate should be one auditable input to a broader safety and governance process - never the sole authority.

## A socially useful certificate should answer six plain questions

**1.** What exactly was compared?
**2.** Which profiles and evaluators were used?
**3.** What stayed the same, and what changed?
**4.** How often is the method wrong, and for whom?
**5.** What happens when confidence is low or profiles disagree?
**6.** Who can audit, correct or override the result?

# 18. A pragmatic pilot plan

The fastest route to a useful deployment is a small, falsifiable pilot. Do not begin by claiming universal geometry. Begin with one transformation, one typed truth and one error budget.

1. Choose one concrete pair of representations, such as scanned form to structured record, CAD model to scan, or camera label to tracked object.
2. Define the typed reference value and the evaluator that decides equality.
3. Declare every profile variable: font, tokenizer, normalization, encoding, padding, device, calibration and units.
4. Build a gold-standard corpus containing positive pairs, negative pairs, hard edge cases and out-of-profile cases.
5. Measure boundary, equivalence and alignment separately. Do not compress them into one score.
6. Issue certificates with preserved features, broken features, reason codes and non-claims.
7. Benchmark p50/p95/p99 latency, throughput, memory and full error rates against a conventional baseline.
8. Set kill criteria before deployment and retain a human-review or safe-fallback path.
9. Repeat after every model, font, schema, firmware or calibration change.

## Three sensible first pilots

| Pilot | Dataset | Primary metrics | Success condition |
|---|---|---|---|
| **Document migration audit** | Versioned sample of forms and database records | semantic agreement, round-trip, hash match, lineage completeness | No silent critical-field change; all exceptions explained. |
| **Font/OCR robustness** | Critical glyphs and documents across fonts, sizes and scans | CER/WER, beta0/beta1/chi, IoU, profile robustness | No forbidden glyph failure; review queue handles uncertainty. |
| **Sensor event gate** | Replayable sensor corpus with calibrated ground truth | false/missed events, p99 latency, guard residual, events/J | Meets hazard budget and beats the baseline at equal error. |

**BEST NEAR-TERM VALUE**

The most immediately practical 3.6.1 product is an auditable transformation certificate and test harness - especially for data migration, OCR/font QA and scientific pipeline provenance.

# 19. When not to use it, and when to stop

## Do not add BEA complexity when a simpler test is enough

- The files must be byte-identical: use a cryptographic hash.
- Only a numeric conversion matters and a normal typed unit test already proves it.
- There is no validated semantic evaluator or reference truth.
- Boundary geometry has no relevance to the application.
- The transformation has no consequence and no audit requirement.

## Kill the application claim when

- The claimed invariant fails under a normal profile change and the application cannot bound the profile.
- False-equivalence or missed-event rates exceed the application budget.
- Calibration drift, quantization or rendering thresholds change event ordering.
- The certificate cannot be reproduced from versioned inputs and definitions.
- A conventional baseline is cheaper, faster or more accurate at equal error.
- The implementation relies on an unsupported leap, such as "same hole count means same topology" or "XOR means encryption."
- The social cost, privacy risk or subgroup error is unacceptable even if technical metrics pass.

**THE SIMPLEST HONEST CONCLUSION**

A failed profile test is useful. It tells you exactly where the claim must become smaller, safer and more specific.

# 20. Final practical conclusion

**LIKE YOU ARE 5**

Version 3.6.1 is a very careful "same or different?" machine. It checks the drawing, the meaning and the computer code separately, and it writes down its homework so another person can check it.

Scientifically, its strongest contribution is a typed certificate for representational transformations. It prevents a local coincidence - one matching number, one repeated mask, one similar sound or one topological feature - from being promoted into total equivalence without the missing evidence.

Pragmatically, the best applications are those where multiple representations of the same item must stay synchronized: document conversion, public and scientific data migration, font/OCR quality, CAD and digital twins, sensor fusion, image-segmentation research, archival integrity and AI robustness audits.

Societally, the value is traceability. A system should be able to say what it compared, how it measured, when it was uncertain, which users or profiles failed, and why it acted or refused to act.

```
Best practical rule: same meaning is not same shape; same shape is not same
code; one matching feature is not total equivalence.
```

## Source basis and evidence labels

| Ref | Source | Use in this guide |
|---|---|---|
| [1] | UGTS-KC 3.6.1 - BEA, Boundary-Equivalence-Alignment. | Primary formal basis: profile, certificate, boundary/equivalence/alignment calculus, 4-font data, exact XOR witness, tests and claims boundary. |
| [2] | UGTS-GN 1.1 GPU-native and physical-device addendum. | Inherited query/event sequence, precision rules, hardware metrics and named software Vulkan benchmark. |
| [3] | Unified Geometric-Topological Substrate UGTS-0. | Typed state, support, compatibility, event, transition, lineage, optional projection and falsification discipline. |
| [4] | Pasted markdown(3).md. | Source motifs for the 3.6.1 distillation. Contextual claims are not treated as established; corrections follow [1]. |

## Evidence label meanings

- Measured in 3.6.1: directly present in the package data or tests.
- Inherited formal capability: specified or measured in the earlier UGTS documents, but not necessarily rebenchmarked for BEA.
- Proposed application: a careful engineering use derived from the formal machinery; requires a new implementation and validation study.
- Not established: a claim explicitly outside the current evidence.

**CLOSING STATEMENT**

Use 3.6.1 as a disciplined comparison and audit layer. Keep every profile explicit, measure the right errors, publish the limits, and let failure narrow the claim before the claim reaches society.

Prepared for Tom Klootwijk - 10-07-1990 - NL200678942. Requester-supplied attribution; not independently verified.